Graphing quadratic functions

1 Consider the function $y = x^2$.

a Complete the following table of values.

x	-3	-2	-1	0	1	2	3
y							

b Plot the points in the table of values on a coordinate plane.

c Hence plot the curve.

d Are the y values ever negative?

e Write down the equation of the axis of symmetry.

f State the minimum y value.

g For every y value greater than 0, how many corresponding x values are there?

2 Consider the function $y = -x^2$.

a Complete the following table of values.

x	-3	-2	-1	0	1	2	3
y							

b Plot the points in the table of values on a coordinate plane.

c Hence plot the curve.

d Are the y values ever positive?

e State the maximum y value.

f Write down the equation of the axis of symmetry.

3 For the following quadratic functions:

i Complete the following table of values.

x	-3	-2	-1	0	1	2	3
y							

ii Graph the function.

a $y = 3x^2$

b $y = -2x^2$

c $y = 3x^2 + 3$

d $y = 3x^2 - 3$

e $y = -2x^2 + 5$

f $y = -2x^2 - 2$

4 For the following quadratic functions:

i Complete the following table of values.

ii Graph the function.

x	-2	-1	0	1	2
y					

a $y = \dfrac{1}{2}x^2$

b $y = -\dfrac{1}{2}x^2$

5 Consider the function $y = (x-2)^2$.

a Complete the following table of values.

x	0	1	2	3	4
y					

b Sketch the graph of the function.

c State the minimum y value.

d State the x value that corresponds to this minimum y value.

e State the coordinates of the vertex.

6 Consider the function $y = -(x-3)^2$.

a Complete the following table of values.

x	1	2	3	4	5
y					

b Sketch the graph of the function.

c State the maximum y value.

d State the x value that corresponds to this maximum y value.

e State the coordinates of the vertex.

7 For the following functions:

i Complete the following table of values.

x	-3	-2	-1	0	1
y					

ii Sketch the graph of the function.

iii State the the minimum y value.

iv State the x value that corresponds to this minimum y value.

v State the coordinates of the vertex.

a $y = (x+1)^2$

b $y = -(x+1)^2$

8 The graph of $y = x^2 + 6$ has no x-intercepts. Is this statement true or false?

9 Consider the equation $y = x^2 + 4$.

a Complete the set of solutions for the given equation.
$A(-3,\square)$, $B(-2,\square)$, $C(-1,\square)$, $D(\square,4)$, $E(1,\square)$, $F(2,\square)$, $G(3,\square)$

b Plot the points C, D and E on the coordinate plane.

c Plot the curve that results from the entire set of solutions for the equation being graphed.

10 Consider the equation $y = x^2 + 6x + 5$.

a Complete the set of solutions for the given equation.
$A(-5, \square)$, $B(-2, \square)$, $C(-6, \square)$, $D(-1, \square)$, $E(0, \square)$, $F(-3, \square)$, $G(-4, \square)$

b Plot the points on the coordinate plane.

c Plot the curve that results from the entire set of solutions for the equation being graphed.

11 Find the maximum value of y for the quadratic function $y = -x^2 + 10x - 25$.

12 Does the curve $y = -x^2 + 3x - 10$ has a minimum value or a maximum value?

13 Consider the curve $y = -2x^2 - 4x + 10$.

a Find the axis of symmetry.

b State the maximum value of y.

c State the coordinates of the vertex.

d Graph the function using the maximum point on the curve.

14 Consider the curve $y = x^2 + 6x + 4$.

a Find the axis of symmetry.

b State the minimum value of y.

c Graph the function using the minimum point on the curve.

15 For the following quadratic functions:

i Find the axis of symmetry.

ii Find the vertex of the parabola .

iii State the y value of the y-intercept of this quadratic function.

iv Sketch the graph of the function.

a $y = 3x^2 - 6x + 8$

b $y = -2x^2 + 4x - 7$

16 Vincent is training for a remote control plane acrobatics competition. He wants to fly the plane along the path of a parabola, and so has chosen the equation $y=3x^2$, where y is the height in meters of the plane from the ground, and x is the horizontal distance in meters of the plane from its starting point.

a Complete the following table of values of the height and distance.

x (m)	0	1	2	3	4
y (m)					

b Plot the shape of the path of the plane.

c State the lowest height of the plane.

d State the x value that corresponds to this minimum y value.

e State the coordinates of the vertex.

17 Consider the quadratic functions, $f(x)$ and $g(x)$ that are shown:
Which function has the smallest minimum value? State its value.

x	-6	-5	-4	-3	-2	-1
$f(x)$	-8	-9	-8	-5	0	7

- $g(x)=x^2-10x+9$